

Artificial Intelligence and the Future Supply of Skills: Evidence from UK University Applications

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Abstract

Educational investments determine the future supply of skills. We examine how generative AI is changing students' decisions about what to study. Linking millions of LinkedIn profiles to UK application records, we measure the AI exposure of university-subject pairs based on the occupations of their past graduates, and follow applications to 1,090 degrees over the 2020–2025 cycles. Human capital theory predicts that students should leave subjects whose prospects are eroded by AI, but we find the opposite. After the release of ChatGPT, applications to a degree one standard deviation higher in exposure grew six percent more than applications to less exposed degrees. The shift appears among men and women and across ethnic groups, while total demand for university places is unchanged. Our labor market analysis shows that the job opportunities for juniors have deteriorated for more exposed degrees. Students are thus moving toward the subjects in which entry-level job opportunities are declining.

Keywords: Generative Artificial Intelligence; Education; Skills; Labor markets

JEL: I23, I26, J23, J24, O33

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1 Introduction

For more than a century, there has been a race between technological change and education (Katz and Murphy, 1992; Card and Lemieux, 2001; Acemoglu, 2002; Goldin and Katz, 2008). During this time, technology has raised the demand for highly educated people more than the increase in university students has increased the supply, leading to increasing graduate premia. Many believe that generative AI will break this pattern (The Economist, 2026; Roose, 2025; Thompson, 2025), as many of the occupations most exposed to large language models are cognitive, white-collar jobs such as programming, accounting, and legal professions (Eloundou et al., 2024; Gmyrek et al., 2025; Tomei and Klein Teeselink, 2026), and recent research shows that demand for these jobs might already be declining (Klein Teeselink and Carey, 2026; Brynjolfsson et al., 2025; Klein Teeselink, 2025).

Whether the pattern in fact breaks, however, depends on more than just the technology itself. Labor markets have two sides, and while a fast-growing body of research examines how AI shifts the demand for skills (see del Rio-Chanona et al., 2025, for a review), we know comparatively little about the supply response. In the past decades, the supply response came from students, whose rising enrollment narrowed but never closed technology’s lead (Goldin and Katz, 2008). This time could be different if the technology substitutes for, rather than complements, the type of skills crafted in university degrees.

In this paper, we study the effect of generative AI on educational choices in the UK. Human capital theory predicts that students should move away from subjects whose expected returns decrease (Becker, 1964). Hence, insofar as the evidence thus far suggests that demand for highly exposed occupations is decreasing, we should expect students to move away from programs that lead to careers with high AI exposure. We test this prediction by studying how university applications to more and less AI-exposed degrees have changed from 2020 to 2025.

We measure educational exposure based on the careers that graduates tend to pursue after their university degree. Using Revelio Labs data containing millions of LinkedIn profiles, we link each combination of university and subject to the occupations its graduates have entered in recent years, and score each degree by the employment-weighted AI exposure of those occupations (Eloundou et al., 2024). We define a degree as a university-by-subject pair (e.g. Economics at King’s College London). The resulting index ranks information technology, statistics, and engineering highest and nursing and education lowest. It also varies considerably across universities within the same subject, because different institutions send graduates into different labor markets. We combine this index with Universities and Colleges Admissions Service (UCAS) records of domestic applications to each university-by-subject from 2020 to 2025 and estimate a difference-in-differences model around the public release of ChatGPT in November 2022, comparing application trends across degrees with different levels of AI exposure.

Our results show that students have moved *toward* subjects with higher AI exposure. In the post-ChatGPT application cycles, a one standard deviation increase in a degree’s exposure is associated with a six percent increase in applications. The effect appears in the first post-release cycle and grows between cohorts. The response is broad, with positive estimates for men and women, for every ethnic group, and in both STEM and non-STEM subjects. Overall demand for university places, by contrast, is unchanged, so students are reallocating between subjects rather than reconsidering the decision to attend university itself.

Whether this reallocation is far-sighted depends on the labor market it anticipates. We therefore complement our university application analysis with an analysis of labor market prospects. Applying the same design, we follow job postings and graduate employment in the occupations that each degree’s graduates enter. While job postings for more exposed degrees show only some evidence of a decline after the release of ChatGPT, the share of junior positions in those occupations has fallen, consistent with Brynjolfsson et al. (2025); Hosseini Maasoum and Lichtinger (2025); Klein Teeselink (2025). Students, in other words, are moving toward subjects in which the entry-level jobs they will likely apply to are disappearing.

Our paper builds on various strands of existing literature. Prior research shows that labor market prospects influence subject choice. Students generally shift toward subjects with better wages and job opportunities (Blom et al., 2021; Long et al., 2015; Acton, 2021). The empirical literature has consistently found, however, that students are only partially responsive to earnings signals, and that the degree of responsiveness depends on the quality of information available to them (de Koning et al., 2025). Importantly, students hold highly inaccurate beliefs about the labor market prospects of different occupations and subjects, particularly for occupations they find attractive (Baker et al., 2018; Betts, 1996; Hastings et al., 2016). Our results extend this evidence to the introduction of generative AI, and show that applications moved toward the subjects whose entry-level prospects were deteriorating, a pattern that is hard to reconcile with students adjusting their choices based on labor demand.

We also build on empirical work that examines the labor market effects of generative AI. Some studies show that employment in exposed occupations has weakened since the release of ChatGPT, especially for junior workers (Brynjolfsson et al., 2025; Hosseini Maasoum and Lichtinger, 2025; Klein Teeselink, 2025; Azar et al., 2025; Klein Teeselink and Carey, 2026), although others do not find such effects (Humlum and Vestergaard, 2025). Studies of earlier generations of AI document reduced employment in exposed work, alongside growth at adopting firms (Acemoglu et al., 2022; Babina et al., 2024; Hample et al., 2025). We extend this evidence from occupations to university degrees, showing that the decline in junior employment carries over to the destination occupations of AI-exposed degrees, and we combine it with university application data to show how the future supply of skills is responding.

While we would expect generative AI-driven labor market effects to play a role

in students’ choice of study, the empirical literature on the subject is still nascent. Two recent papers measure how exposed higher education is to AI. Timmerman (2025) maps college majors to occupations and finds that STEM subjects are more highly exposed than non-STEM subjects. Light (2026) measures exposure inside the classroom instead, scoring the task content of coursework at more than 1,000 US universities against LLM capabilities. Both papers document that exposure is concentrated in writing- and data-intensive subjects. Light (2026) additionally shows that instructors are partially adapting and that US computer science enrollment share fell in 2025–26 for the first time in two decades. However, neither paper estimates how students respond to exposure. Prior research on physical automation shows that technology shocks move aggregate education demand (for instance, Di Giacomo and Lerch (2026) find that every additional industrial robot increases local university enrollment by four students), yet evidence on how these types of shocks change students’ choice of degree program remains limited. We fill this gap by providing some of the first evidence on what students choose to study in response to AI exposure.

2 Methodology

2.1 Data

Our analysis combines LinkedIn profile data from Revelio Labs, university application data from UCAS, and occupation-level AI exposure scores from Eloundou et al. (2024).¹

Our labor market data come from Revelio Labs, which aggregates and builds labor force datasets from public workforce data. Our sample consists of the universe of LinkedIn profiles of UK workers. The data contain each worker’s full work history alongside their education records. For each spell of employment, we observe the start and end dates, the job title, the (imputed) salary, and standardized occupation codes. The educational record reports the degree level, subject, university name and country, and graduation date for each credential listed on a profile. Revelio standardizes the reported subject of each credential into 17 broad categories, such as engineering, information technology, and economics.

From these profiles, we construct an education-level AI exposure index, which measures, for each undergraduate subject at each university, how exposed its graduates’ careers are to AI. We henceforth refer to a university-by-subject pair as a ‘degree’. We build the index from all bachelor’s graduates of UK universities in the

¹Eloundou et al. (2024) assess, for each task in the O*NET database, whether access to a large language model can reduce the time required to perform the task by at least half at equal quality. Their α score captures tasks that an LLM can affect on its own, while the broader β score adds tasks that become exposed once additional software is built on top of the model.

2002 to 2022 graduation cohorts. Because our Revelio extract contains only positions active on or after January 2021, we observe the jobs these graduates hold between January 2021 and October 2022, including jobs that began years earlier. The index therefore captures graduates’ occupational destinations on the eve of generative AI, observed between zero and twenty years into their careers. We construct an alternative variant that uses only the first job after graduation. We proceed in two steps. First, for each graduate, we average the exposure scores of their post-graduation jobs, weighting each job by the number of months the graduate held it after graduation. Second, we average these graduate-level exposures across all graduates of the same subject at the same university, counting each graduate once and dropping degrees with fewer than ten graduates. We only use labor market information from before the release of ChatGPT in constructing the index to avoid contamination from LLMs. We end up with an exposure score at the subject-university level, containing variation in exposure both between subjects and within subjects across universities.

Our main university application data come from UCAS, which operates the admissions process for UK undergraduate degrees. We use its provider-level microdata, which contain the yearly number of applications per subject per university between 2020 and 2025. We define a cycle as the year in which applicants would start their degree. Hence, all applications for a September 2023 start are part of the 2023 cycle. We focus on domestic applicants (domiciled in England, Scotland, Wales, or Northern Ireland) to avoid confounding from international student flows, whose education decisions may be influenced by other factors such as Brexit. Alongside this microdata, we use the public UCAS end-of-cycle release, which reports national application totals by subject group and by demographic characteristics (gender, nation of domicile, and ethnicity). This dataset supports our demographic heterogeneity analysis.

To construct our final dataset, we merge our education-level AI exposure data with the UCAS application data at the university $\times$ subject level. Because Revelio records university names as free text, we first match each name to its UCAS provider counterpart using a similarity score, and we review uncertain matches by hand.² The confirmed matches cover 97 percent of the graduate records in our sample. UCAS classifies subjects using the Higher Education Classification of Subjects (HECoS) Common Aggregation Hierarchy, from CAH1, the broadest level, to CAH3, the most granular. For example, CAH1 groups subjects into broad categories such as “Business and Management”, while CAH3 distinguishes “Accounting”, “Marketing”, and “Finance”. Revelio classifies each graduate’s subject into one of 17 broad categories. We map these categories to 12 of the 21 CAH1 subject groups and conduct our main

²We match normalized names on token-set similarity, accepting matches that score above 0.85 and hand-reviewing lower-scoring candidates. The confirmed crosswalk links 255 Revelio university names to 225 UCAS providers.

analysis at the provider $\times$ CAH1 level. The remaining nine CAH1 groups, such as psychology, languages, and the creative arts, have no defensible Revelio counterpart and are excluded. As a robustness check, we re-estimate our models at the more granular CAH3 level, mapping each Revelio category to the closest CAH3 subject groups (Section 3.1). Appendix A1 contains additional details on the matching procedure.

We focus on all degrees with at least ten applications in 2022 and with an exposure index based on a minimum of ten LinkedIn profiles. We exclude degrees with zero applications in any cycle. Our final estimation panel consists of 1,090 university-subject pairs across six application cycles, yielding a total of 6,540 observations. Our main outcome of interest is the log number of applications to each degree in each cycle.

Table A3 shows summary statistics across broad subject categories. The most exposed subjects are computing (0.59), mathematical sciences (0.55), and engineering (0.55), and the least exposed are subjects allied to medicine (0.38), education (0.40), and medicine (0.43). Exposure also varies across universities that offer the same subject, with a within-subject standard deviation between 0.01 for law and 0.05 for computing and engineering. The final column reports the raw change in applications after the release of ChatGPT. Applications grew by 22 percent in computing and by 14 percent in engineering, and they fell by 17 percent in education and by 11 percent in subjects allied to medicine. Figure A1 shows total undergraduate applicant numbers over the 2018/19 to 2024/25 academic years, which remained broadly stable.

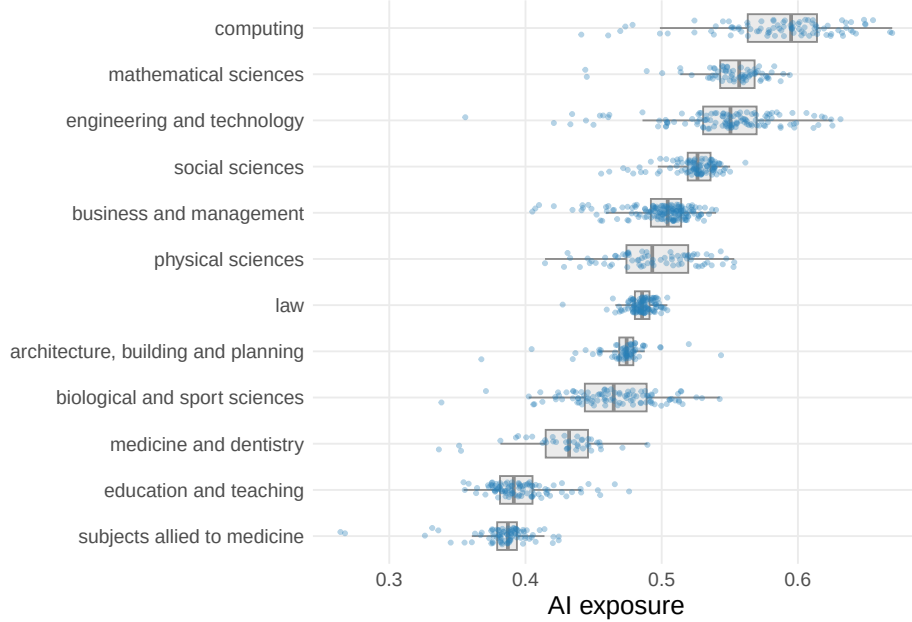
Figure 1 plots the exposure index for every university-subject pair in the estimation sample. The within-subject variation is large enough to make many subjects overlap. For example, exposure in computing degrees ranges from 0.44 to 0.67, and the least exposed computing degrees are substantially less exposed than the median physical sciences degree, which sits in the middle of the subject exposure ranking. Law has by far the narrowest exposure distribution, suggesting that law graduates enter occupations with similar levels of AI exposure independent of where they study.

2.2 Estimation Strategy

To estimate the effect of education-level AI exposure on applications, we use a difference-in-differences design that compares degrees with higher and lower AI exposure before and after the introduction of ChatGPT in November 2022. We select 2022 as the base year because it represents the last application cycle in which all application decisions were made before generative AI became widely available. Our panel covers three pre-treatment cycles (2020–2022) and three post-treatment cycles (2023–2025).

We estimate the following specification:

Figure 1: Exposure by subject, across universities



Notes: This figure plots the AI exposure index of every university-subject pair in the estimation sample. Each point is one pair, and boxes summarize the distribution of exposure within each subject. Subjects are ordered by their median exposure. The index is the all-jobs β exposure measure described in Section 2.1.

$$\log(\text{Applications}_{pst}) = \theta \cdot (\text{Exposure}_{ps} \times \text{Post}_t) + \alpha_{ps} + \delta_t + \varepsilon_{pst} \quad (1)$$

where $\text{Applications}_{pst}$ denotes total domestic UCAS applications to university p in subject s during application cycle t . Exposure_{ps} is the time-invariant AI exposure index of the degree, standardized across degrees, so that θ measures the effect of a one standard deviation increase in exposure. Post_t is an indicator equal to 1 for post-ChatGPT cycles ($t \geq 2023$). α_{ps} are university-by-subject (‘degree’) fixed effects, which control for all time-invariant differences across degrees, such as selectivity and baseline popularity. Hence, we only use within-degree variation in applications over time. δ_t are year fixed effects, which control for general trends in application volumes, as well as shocks such as demographic changes and economic conditions that affect all degrees equally. In stricter specifications, we add subject-by-year fixed effects (δ_{st}). Standard errors are two-way clustered by university and subject.

We also estimate the analogous event study specification:

$$\log(\text{Applications}_{pst}) = \sum_{\tau \neq 2022} \theta_{\tau} \cdot (\text{Exposure}_{ps} \times \mathbf{1}[t = \tau]) + \alpha_{ps} + \delta_t + \varepsilon_{pst}, \quad (2)$$

where θ_τ traces the effect of exposure on applications in cycle τ relative to 2022, the final pre-treatment cycle.

Our identification strategy relies on the parallel trends assumption, which holds that absent the emergence of large language models, application trends across subjects with different levels of AI exposure would have evolved similarly, conditional on our fixed effects structure. Three features of the design support this assumption. First, the exposure index is constructed from labor market information through October 2022, so changes in applications cannot feed back into the measure of exposure. Second, the timing of the treatment came from outside the UK admissions system. OpenAI released ChatGPT in the middle of the 2023 application cycle, and neither applicants nor universities could have anticipated the release. Third, our event study estimates show that more and less exposed degrees evolved similarly before the introduction of ChatGPT (Figure 2).

We perform a number of robustness checks to test the validity of our findings. We re-estimate our analysis using alternative subject mappings (estimating at the more granular CAH3 subject level). We also examine effects on total instead of log applications and re-run the analysis while weighting degrees by their 2022 application count. We additionally consider alternative exposure measures such as the α measure from Eloundou et al. (2024), the measure of Gmyrek et al. (2025), and a version of our exposure measure based only on graduates' first jobs after finishing their degree. Finally, we perform a leave-one-out analysis, where we iteratively remove each subject from the analysis.

Finally, we supplement our application analysis with an examination of labor market prospects across degrees. To do so, we construct a monthly panel of job postings and employment associated with each degree from January 2021 to October 2025. To attribute jobs to degrees, we start from the distribution of occupations that each degree's graduates held in October 2022, the last month before treatment, and keep degrees with at least 50 active positions in that month. Each degree then receives a share of every occupation's monthly totals, equal to its graduates' share of that occupation's UK positions in October 2022. For example, if economics graduates held 20 percent of all investment banking jobs in October 2022, we attribute 20 percent of investment banking job postings and employment to economics in every month of our window. Our outcomes are the log number of job postings facing each degree, and the junior share of its employment.³ We estimate the same event study design as for applications, regressing each degree's change in the outcome relative to October 2022 on its standardized exposure, with shift-share standard errors (Adão et al., 2019).⁴

³The junior share is defined as the fraction of employment in the lowest two seniority categories of Revelio Labs, which constitute entry-level and junior positions.

⁴We cluster the occupation shocks by SOC minor group and, more conservatively, by SOC major group.

Table 1: Effect of AI exposure on UCAS applications

Model:	(1)	(2)
AI exposure x post	0.0596*** (0.0132)	0.0354** (0.0144)
<i>Fixed-effects</i>		
Provider x Subject	Yes	Yes
Year	Yes	
Subject x Year		Yes
<i>Fit statistics</i>		
Observations	6,540	6,540
R ²	0.979	0.980
Within Adjusted R ²	0.023	0.002

Notes: This table reports difference-in-differences estimates of the effect of AI exposure on log applications, estimated from Equation (1). The treatment, AI exposure x post, is the standardized all-jobs β exposure index interacted with a post-ChatGPT indicator equal to one from the 2023 cycle onward. Column 1 includes provider-by-subject and year fixed effects. Column 2 adds subject-by-year fixed effects, so identification comes only from differences in exposure across providers within a subject. Standard errors, in parentheses, are two-way clustered by provider and subject. Asterisks denote significance at the 0.01 (***), 0.05 (**), and 0.1 (*) level.

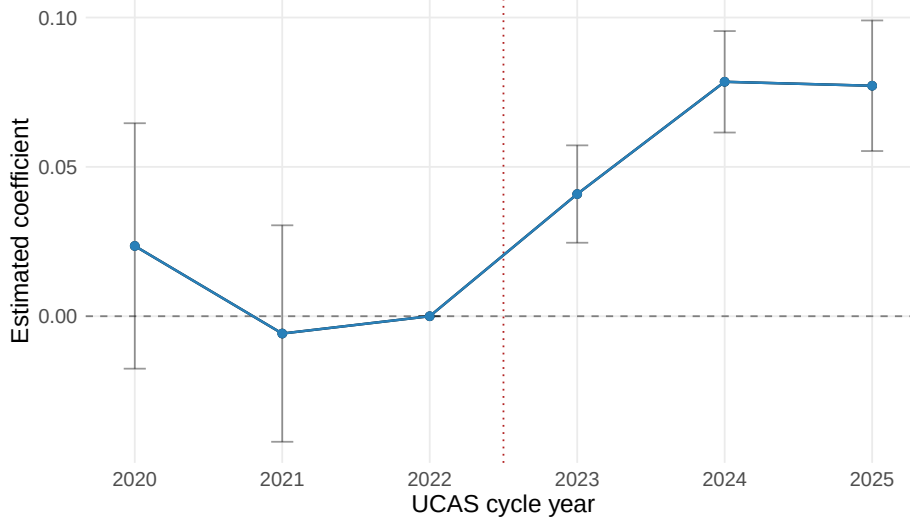
3 Results

We start by analyzing the effect of AI exposure on UK university applications by estimating Equation (1). Table 1 reports the results. After the release of ChatGPT, applications to a degree one standard deviation higher in AI exposure grew by 0.0596 log points more than applications to less exposed degrees (column 1), an increase of roughly six percent, or about 60 additional applications per cycle for the median degree. The effect remains positive and significant when we add subject-by-year fixed effects that control for any subject-level shock in a given cycle. The estimated effect is smaller than in our main specification, likely because identification now comes only from differences in exposure across universities within the same subject, thus removing any overall subject-level effects of AI exposure.

Figure 2 shows the event study plot. In the years before the introduction of ChatGPT, high- and low-exposure degrees follow similar trajectories in terms of application growth. After ChatGPT launches, however, the trajectories begin to diverge, and applications to more exposed degrees start to increase. In 2023, applications increase by four percent for each additional standard deviation of AI exposure. By 2024, this effect grows to eight percent. The effect levels off by 2025.

These changes in applications are informative about the future supply of skills.

Figure 2: Effect of AI exposure on applications, event study



Notes: This figure plots event study estimates of the effect of AI exposure on log applications. Coefficients are estimated from Equation (2) and are shown relative to the 2022 cycle, the final pre-treatment cycle. The regression includes university-by-subject and year fixed effects. Standard errors are two-way clustered by university and subject. The vertical line marks November 2022, when ChatGPT was publicly released. Error bars denote 95 percent confidence intervals.

UK students apply to a specific subject and rarely switch after entry, so the composition of applications strongly correlates with the composition of graduates three to four years later. The cohorts that applied in the 2023 to 2025 cycles will reach the labor market between 2026 and 2028, and if acceptance and completion rates hold up, these results imply an incoming increase in the supply of the types of skills that are typically required in more AI-exposed occupations.

3.1 Robustness Checks

Our main application result does not hinge on exactly how we measure exposure or how finely we classify degrees. Figure A2 shows the results for analyses where we replace the β exposure index with the narrower α index of Eloundou et al. (2024), the ISCO-2025 index of Gmyrek et al. (2025), and the first-job variant of our main exposure index, which uses only each graduate’s first observed job after graduation. The results are qualitatively and quantitatively similar to our main results. When we map the 17 Revelio fields to 87 CAH3 subjects, expanding the panel to 3,237 degrees, all conclusions also remain the same (Figure A3).

The results are also robust to other design choices. With raw application counts as the outcome instead of log applications, a one standard deviation increase in

exposure is associated with 134 additional applications per cycle after the release of ChatGPT (Figure A4). Weighting each degree by its 2022 application count raises our main estimate to 0.0832 log points (Figure A5). In other words, the reallocation is, if anything, stronger among the large degrees that receive the most applications. Finally, no single subject drives the result. Dropping each of the 12 subject groups in turn moves the estimate between 0.0457 and 0.0653 (all significant; Figure A6). The lower bound comes from excluding computing, and even there the effect is more than three quarters of our headline magnitude. Table A4 contains the static diff-in-diff estimates for all robustness checks. Taken together, the robustness checks show that our results do not depend on specific modeling choices or data restrictions.

3.2 Heterogeneity Tests

We next ask which degrees and institutions are driving our results. Using the main data, we estimate the effect separately by subject type (STEM or non-STEM),⁵ university prestige (Russell Group or not),⁶ and university size (terciles of pre-treatment application volume). We also explore heterogeneity by gender and ethnicity, but these splits are based on a much smaller sample of publicly available UCAS data that measures applications aggregated at the subject level rather than the university-by-subject level.^{7,8}

Figure A7 reports the heterogeneity results for our main sample. The point estimate is positive in both STEM and non-STEM subject groups, though the non-STEM estimate is imprecise. The effect is larger in STEM, but the difference between the two estimates is not statistically significant. The effect is about twice as strong at Russell Group universities (0.0917) as elsewhere (0.0471), but again this difference is insignificant.

When we consider heterogeneity by size, we find that the effect is only present in medium-sized and large universities. For small universities, the effect is negative and insignificant. In other words, the reallocation toward AI-exposed degrees is concentrated in large institutions that receive the most applications.

Turning to demographics, Figure A8 shows positive estimates for men and women, as well as for every ethnic group. In the public data, applications to subjects one standard deviation higher in exposure grow by 0.0782 log points (roughly eight percent) more after the release of ChatGPT, in line with our microdata results. The

⁵The STEM group comprises medicine and dentistry, subjects allied to medicine, biological and sport sciences, physical sciences, mathematical sciences, engineering and technology, computing, and architecture, building and planning; the non-STEM group comprises social sciences, law, business and management, and education and teaching.

⁶The Russell Group is an association of 24 research-intensive universities, including Oxford and Cambridge, and membership is the standard marker of institutional prestige in UK higher education.

⁷Figure A9 shows that our main results replicate on this sample.

⁸For this analysis, we include subject and year fixed effects.

effect is somewhat larger for women (0.0945) than for men (0.0812), but the two confidence intervals overlap almost entirely. The five ethnic groups show a similar pattern, as every estimate is positive, ranging from 0.0502 for Asian applicants to 0.0959 for Black applicants, and none are significantly different from each other.

Overall, applications shift toward exposed subjects in almost every subsample we examine, with small universities as the only exception. Our results thus indicate that the composition of the future skill supply is broadly shifting toward AI-exposed subjects. In the next subsection, we examine how labor market prospects of these degrees have changed since the introduction of ChatGPT.

3.3 Labor Market Results

The shift toward more AI-exposed subjects could be rationalized if the labor market prospects of those subjects have improved as well, because an improvement would signal rising demand for the skills students develop in those degrees. To test this, we examine how two measures of labor market prospects, namely new job postings and the junior share of employment in each degree’s destination occupations, have evolved since the introduction of ChatGPT. For methodological details, see Section 2.2.

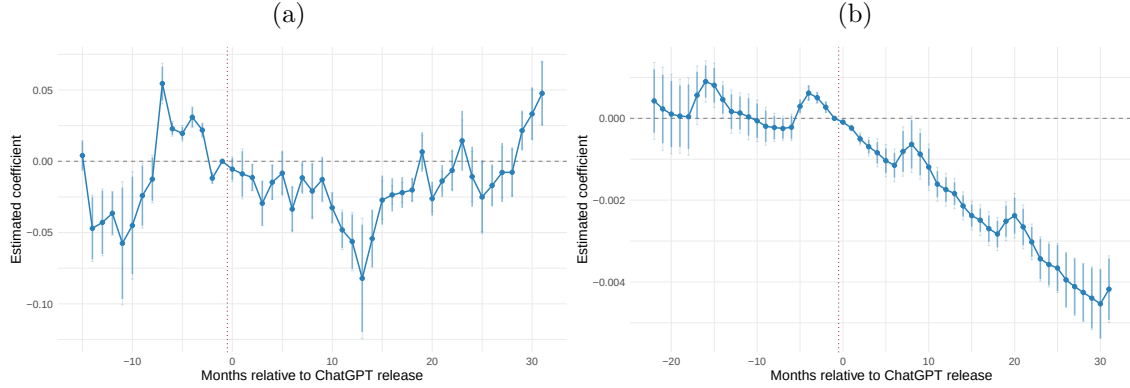
Figure 3 shows the event study plots for job postings and the share of junior employment. The results suggest that labor market prospects of graduates from highly exposed degrees have deteriorated since the introduction of ChatGPT. Averaged over the post-treatment period, postings fall by 0.8 percent and the share of junior employment declines by 0.25 percentage points per standard deviation of exposure, although the former is only statistically significant in some specifications (Table A5). This deterioration is consistent with broader UK labor market evidence from Klein Teeselink (2025), and suggests that university applicants either do not observe this deterioration or do not weigh it when choosing their degrees, possibilities we return to in Section 4.

4 Conclusion

This paper asks whether students change what they study in response to the public introduction of generative AI. The answer is of first-order importance, because educational choices today determine a country’s supply of skills in the future, and if skill investments do not track the skills demanded in the labor market, countries may end up with severe skill mismatches, leading to shortages in some skills, surpluses in others, and slower productivity growth.

To measure how exposed each degree is to generative AI, we construct a new AI exposure measure for UK university degrees based on the occupations their graduates enter. For every subject in each university, we average the exposure scores of

Figure 3: Effect of AI exposure on job postings and the junior share of employment, event study



Notes: This figure plots monthly event study estimates of the effect of AI exposure on log UK job postings (panel A) and on the junior share of employment (panel B) facing each degree's graduates. The junior share is the fraction of positions in the two lowest Revelio seniority levels, which cover entry-level and junior roles. Postings and employment are assigned to degrees using each degree's fixed share of every occupation's positions in October 2022, the final pre-treatment month and the reference period. The vertical line marks November 2022, when ChatGPT was publicly released. Error bars denote 95 percent confidence intervals. The dark inner bars cluster the occupation shocks by SOC minor group and the light outer bars by SOC major group.

its graduates' occupations, where an occupation's score is the share of its tasks that large language models could substantially speed up (Eloundou et al., 2024). We then compare applications to more and less exposed degrees before and after the release of ChatGPT. Our results show that students moved toward, rather than away from, AI-exposed subjects. After the release of ChatGPT, applications to highly exposed degrees rose by roughly six percent compared to applications to less exposed degrees. The response is broad, with positive estimates for men and women, for every ethnic group, and in both STEM and non-STEM subjects. We furthermore show that labor market prospects for highly exposed degrees have deteriorated since the introduction of ChatGPT, as the junior share of employment has fallen by 0.25 percentage points per standard deviation of exposure. In other words, applicants are increasingly entering degrees with weakening entry-level opportunities.

One possible interpretation is that students simply do not see the deterioration. Students hold systematically optimistic beliefs about the subjects they find attractive (Baker et al., 2018; de Koning et al., 2025), and changes in hiring trends for junior roles may not reach applicants quickly enough, if at all. A second possibility is that students see AI as an opportunity rather than a threat. A computer science degree, for example, may now attract applicants who want to build or work with the

technology, and AI tools may lower the barriers into high-paying technical careers. The labor market evidence, however, gives no sign that prospects in these subjects have improved, at least so far. A third is that other shocks pushed in the same direction. The rise of remote work and the surge in inflation were still unfolding when ChatGPT arrived, and students worried about financial security may have moved toward subjects they associate with high pay, which happen to be more exposed to AI as well. Both shocks, however, predate ChatGPT by one to three years, whereas applications only begin to shift toward exposed subjects in the 2023 cycle, which, in terms of timing, aligns with the introduction of ChatGPT.

We acknowledge several limitations to our analysis. While the timing of the effects narrows the set of alternative explanations, we cannot rule out that some shocks that arrived at the same time as ChatGPT are correlated with exposure, and our estimates may partly reflect such shocks rather than the technology itself. The exposure index has limits of its own. It captures the share of tasks that today’s models could substantially speed up, but not the value of those tasks within a job, how occupations will change over a graduate’s career, or whether exposure leads to actual automation. Moreover, AI may increase the optionality of various degrees. Political science graduates may reasonably become software engineers, as the barrier to coding drops thanks to emerging AI tools. Finally, applications measure demand from students, not supply from universities. If universities increased capacity or intensified recruitment in exposed subjects over the same period, part of the response we attribute to students may reflect these institutional choices instead.

Our findings have important implications for policymakers. Students are real-locating toward subjects with weakening entry-level demand, and if these trends continue, the gap between the skills graduates supply and the skills employers demand will widen over the coming years. To counter this mismatch, policymakers could first improve the information on which students base their choices. Beliefs about labor market prospects are often outdated and overly optimistic, but experimental evidence shows that accurate labor market information corrects such beliefs (de Koning et al., 2025). Information campaigns and career guidance could therefore give students current evidence on entry-level hiring in the subjects they are considering. Second, policymakers could invest in retraining. A degree chosen today reaches the labor market only years later, so even well-informed students will sometimes graduate into subjects that weakened along the way. Conversion courses and other forms of retraining can absorb part of this mismatch, and they will become more important as AI continues to shift the demand for skills. The evidence on retraining is mixed, however, and most evaluations find only modest gains on average (Jacobs and Canedy, 2026; Roodman and Massenkoff, 2026). Insofar as policymakers want to pursue this route, retraining works best when it screens applicants intensively, involves employers in deciding what to teach, and tracks local demand for skills (Roodman and Massenkoff, 2026).

We set out to test whether generative AI pushes students away from AI-exposed

subjects. So far it has done the opposite. The coming application cycles, in the UK and elsewhere, will show whether that holds once the labor market consequences of AI become easier to observe.

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A1 Subject Mapping

To merge the exposure index with the UCAS application data, we require a crosswalk between Revelio Labs and UCAS subject classifications. Revelio standardizes each graduate’s subject into one of 17 broad fields of study, whereas UCAS classifies applications with the Higher Education Classification of Subjects (HECoS) Common Aggregation Hierarchy, which nests 166 detailed CAH3 subjects within 21 broad CAH1 groups. CAH1 divides subjects into broad categories such as “computing”, while CAH3 distinguishes “software engineering” and “computer science”. Because we build the exposure index at the university $\times$ Revelio-field level, we must map the 17 fields onto one level of this hierarchy. We choose CAH1 for the main analysis because the Revelio fields align closely with the CAH1 groups, so most mappings are one-to-one. At CAH3, a single field would have to stand in for many narrower subjects.

Table A1 reports the Revelio to CAH1 mapping. Twelve of the 21 CAH1 groups have a Revelio counterpart, and together they account for 73.5 percent of UCAS applications over the 2020–2025 cycles. The nine unmatched groups, among them psychology, the creative arts, and languages, account for the remaining 26.5 percent and are excluded. Nine of the twelve retained groups draw on exactly one field. The other three pool closely related fields: physical sciences combines physics and chemistry; mathematical sciences combines mathematics and statistics; and business and management combines accounting, business, finance, and marketing. For these pooled groups, we compute exposure as the mean over the pooled fields’ graduates, so each field contributes in proportion to its graduate count.

Table A1: Mappings from Revelio Labs fields of study to HECoS CAH1 subject groups

Revelio field	CAH1 subject group
Medicine	Medicine and dentistry (CAH01)
Nursing	Subjects allied to medicine (CAH02)
Biology	Biological and sport sciences (CAH03)
Physics	Physical sciences (CAH07)
Chemistry	
Mathematics	Mathematical sciences (CAH09)
Statistics	
Engineering	Engineering and technology (CAH10)
Information technology	Computing (CAH11)
Architecture	Architecture, building and planning (CAH13)
Economics	Social sciences (CAH15)
Law	Law (CAH16)
Accounting	Business and management (CAH17)
Business	
Marketing	
Finance	
Education	Education and teaching (CAH22)

Notes. The 17 Revelio Labs fields of study map onto 12 of the 21 HECoS CAH1 subject groups, covering 73.5% of UK undergraduate application volume (2020–2025 UCAS cycles); the remaining 26.5% falls in the 9 CAH1 groups with no Revelio field at all (e.g. psychology, design and creative arts, language and area studies). Physical sciences, mathematical sciences, and business and management pool multiple Revelio fields into one CAH1 group; the pooled exposure is a graduate-weighted (not field-averaged) mean across the pooled fields.

For the robustness check in Section 3.1, we instead map the 17 fields onto the 166 CAH3 subjects. Table A2 reports this mapping. We assign each field’s index to the CAH3 subjects whose graduates it plausibly describes, and we tag each assignment as an exact match, a sub-discipline, or a proxy. Coverage is lower than at CAH1 because the finer mapping is more conservative. Where CAH1 lets economics stand in for all of social sciences, CAH3 keeps only economics itself. In total, 87 of the 166 subjects receive an index, covering 61.7 percent of applications. Applying the same sample filters as the main analysis leaves 3,237 university-subject pairs, covering 56.3 percent of applications.

Table A2: Mappings from Revelio Labs fields of study to HECoS CAH3 subjects

Revelio field	CAH3 subject(s)	HECoS code(s)
Medicine	Medicine (non-specific); medicine by specialism [exact]; medical sciences (non-specific) [sub]; dentistry [proxy]	(CAH01-01-02)–(CAH01-01-03) [exact]; (CAH01-01-01) [sub]; (CAH01-01-04) [proxy]
Nursing	Nursing (non-specific); adult nursing; community nursing; children's nursing; dental nursing; mental health nursing; learning disabilities nursing; others in nursing [exact]; midwifery [sub]; pharmacy; medical technology; healthcare science (non-specific); health sciences (non-specific); nutrition and dietetics; ophthalmics; environmental and public health; physiotherapy; complementary and alternative medicine; counselling, psychotherapy and occupational therapy [proxy]	(CAH02-04-01)–(CAH02-04-03), (CAH02-04-05)–(CAH02-04-09) [exact]; (CAH02-04-04) [sub]; (CAH02-02-03), (CAH02-05-01)–(CAH02-05-02), (CAH02-06-01)–(CAH02-06-07) [proxy]
Biology	Biosciences (non-specific); biology (non-specific) [exact]; pharmacology; toxicology; biomedical sciences (non-specific); anatomy, physiology and pathology; ecology and environmental biology; microbiology and cell science; plant sciences; zoology; genetics; molecular biology, biophysics and biochemistry; others in biosciences; biotechnology [sub]; environmental sciences [proxy]	(CAH03-01-01)–(CAH03-01-02) [exact]; (CAH02-02-01)–(CAH02-02-02), (CAH02-05-03)–(CAH02-05-04), (CAH03-01-03)–(CAH03-01-08), (CAH03-01-10), (CAH10-03-05) [sub]; (CAH26-01-04) [proxy]
Physics	Physics [exact]; astronomy [sub]; physical sciences (non-specific) [proxy]	(CAH07-01-01) [exact]; (CAH07-01-02) [sub]; (CAH07-04-01) [proxy]
Chemistry	Chemistry [exact]	(CAH07-02-01) [exact]
Mathematics	Mathematics [exact]; operational research [sub]	(CAH09-01-01) [exact]; (CAH09-01-02) [sub]
Statistics	Statistics [exact]	(CAH09-01-03) [exact]
Engineering	Engineering (non-specific) [exact]; mechanical engineering; production and manufacturing engineering; aeronautical and aerospace engineering; naval architecture; bioengineering, medical and biomedical engineering; civil engineering; electrical and electronic engineering; chemical, process and energy engineering; others in engineering [sub]; minerals technology; materials technology; polymers and textiles; maritime technology; others in technology; materials science [proxy]	(CAH10-01-01) [exact]; (CAH10-01-02)–(CAH10-01-10) [sub]; (CAH10-03-01)–(CAH10-03-04), (CAH10-03-06)–(CAH10-03-07) [proxy]
Information technology	Computer science; information technology [exact]; information systems; software engineering; artificial intelligence; computer games and animation; business computing; others in computing [sub]	(CAH11-01-01)–(CAH11-01-02) [exact]; (CAH11-01-03)–(CAH11-01-08) [sub]
Architecture	Architecture [exact]; landscape design [sub]; building; planning (urban, rural and regional) [proxy]	(CAH13-01-01) [exact]; (CAH13-01-03) [sub]; (CAH13-01-02), (CAH13-01-04) [proxy]
Economics	Economics [exact]	(CAH15-02-01) [exact]
Law	Law [exact]	(CAH16-01-01) [exact]
Accounting	Accounting [exact]	(CAH17-01-08) [exact]
Business	Business and management (non-specific); business studies [exact]; management studies; human resource management; others in business and management [sub]; tourism, transport and travel [proxy]	(CAH17-01-01)–(CAH17-01-02) [exact]; (CAH17-01-04)–(CAH17-01-05), (CAH17-01-09) [sub]; (CAH17-01-06) [proxy]
Marketing	Marketing [exact]; publicity studies [proxy]	(CAH17-01-03) [exact]; (CAH24-01-02) [proxy]
Finance	Finance [exact]	(CAH17-01-07) [exact]
Education	Education [exact]; teacher training [sub]	(CAH22-01-01) [exact]; (CAH22-01-02) [sub]

Notes. Each cell groups codes by match quality: *exact* (same subject as the Revelio field), *sub* (a sub-discipline of it), *proxy* (the nearest defensible graduate pool, used as coarser stand-in). Overall, 61.7% of UK undergraduate application volume (2020–2025 UCAS cycles) falls in a mapped CAH3 subject; 56.3% is retained in the final provider $\times$ CAH3 $\times$ year estimation panel (3237 cells) used for this robustness check, after the same ≥ 10 -applications / ≥ 10 -profiles / no-zero-cycle filters as the CAH1 headline panel.

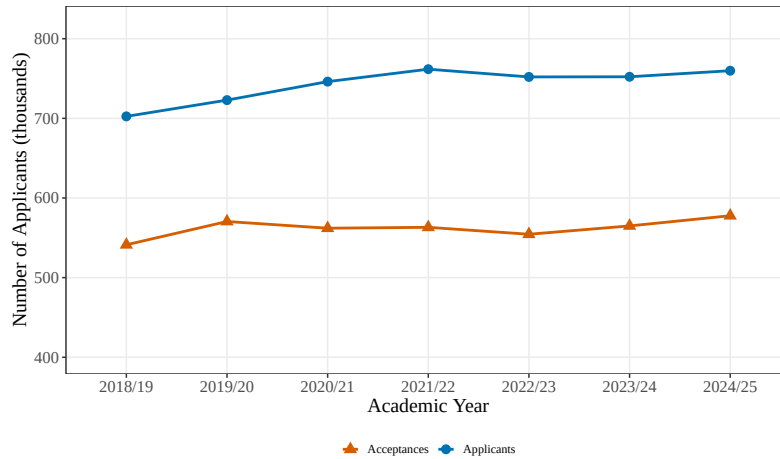
A2 Additional Figures and Tables

Table A3: AI exposure and applications by subject (CAH1), estimation sample

Subject	Beta exposure	SD across providers	Provider cells	Median profiles	Applications (000s)	Growth post vs pre (%)
Computing	0.586	0.046	90	39.5	614	+22
Mathematical sciences	0.551	0.027	67	480.0	224	+10
Engineering and technology	0.546	0.045	118	1197.0	799	+14
Social sciences	0.523	0.019	98	266.5	1377	-5
Business and management	0.498	0.025	154	2092.5	1548	+6
Physical sciences	0.494	0.033	71	293.0	354	+2
Law	0.485	0.010	111	804.0	700	+13
Architecture, building and planning	0.472	0.022	68	344.0	235	+5
Biological and sport sciences	0.465	0.034	98	200.0	774	-3
Medicine and dentistry	0.426	0.031	39	271.0	530	-3
Education and teaching	0.397	0.023	92	362.5	393	-17
Subjects allied to medicine	0.383	0.025	84	202.5	1942	-11

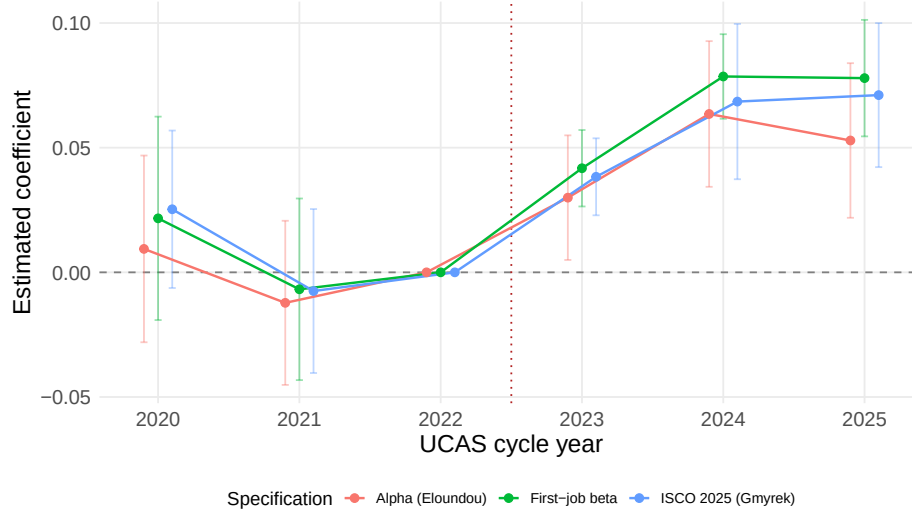
Notes: This table reports the AI exposure index and applications by subject group in the estimation sample. Each row is one CAH1 subject group, ordered by its mean all-jobs β exposure (Eloundou et al., 2024). *SD across providers* is the standard deviation of exposure across universities offering the subject. *Provider cells* is the number of university-subject pairs, and *Median profiles* is the median number of LinkedIn profiles behind a pair's exposure index. *Applications* is the total number of applications over the sample period, in thousands. *Growth* compares total applications in the post-ChatGPT cycles (2023–2025) to the pre-treatment cycles (2020–2022).

Figure A1: Total UK undergraduate applicants and acceptances, 2018/19–2024/25



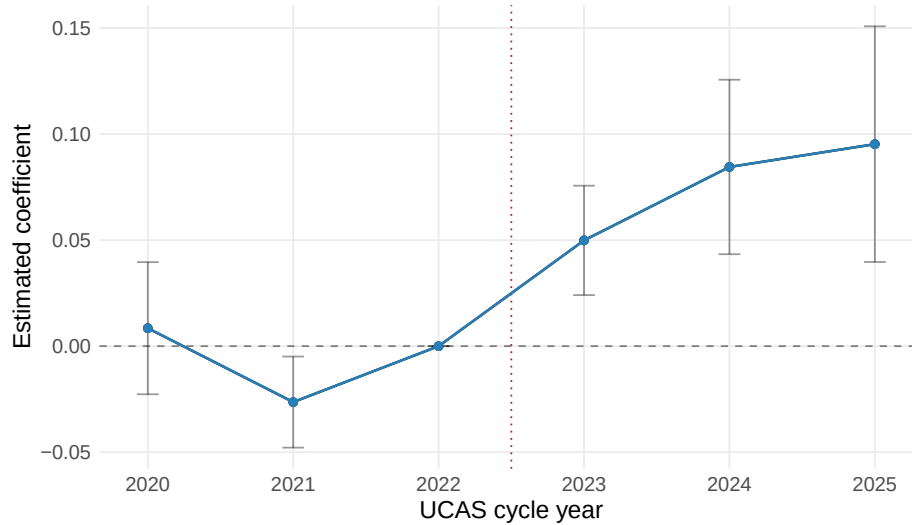
Notes: This figure plots the total number of UK undergraduate applicants and acceptances, in thousands, over the 2018/19–2024/25 academic years. Data come from the public UCAS end-of-cycle release.

Figure A2: Event study, alternative exposure measures



Notes: This figure plots event study estimates from Equation (2), replacing the all-jobs β exposure index with the α index of Eloundou et al. (2024), the ISCO-2025 index of Gmyrek et al. (2025), and a first-job variant that uses only each graduate's first observed job after graduation. All indices are standardized. All other definitions are as in Figure 2.

Figure A3: Event study, CAH3 subject level



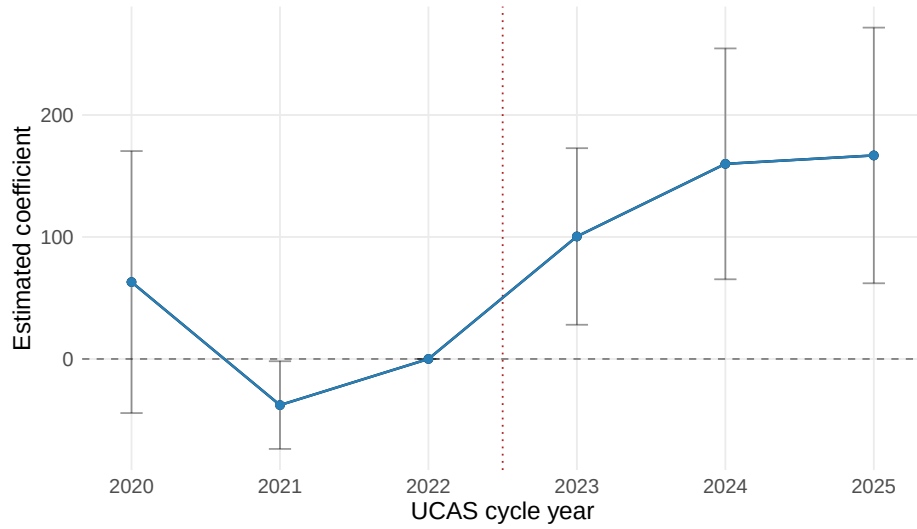
Notes: This figure plots event study estimates from Equation (2), estimated at the more detailed CAH3 subject level, with standard errors two-way clustered by university and Revelio category. All other definitions are as in Figure 2.

Table A4: Robustness of the static difference-in-differences estimate

Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
All-jobs beta exposure (SD) x Post	0.0596*** (0.0132)				0.0825*** (0.0155)	134.0*** (32.44)	0.0832*** (0.0123)
All-jobs alpha exposure (SD) x Post		0.0497*** (0.0153)					
All-jobs ISCO exposure (SD) x Post			0.0534** (0.0188)				
First-job beta exposure (SD) x Post				0.0611*** (0.0130)			
<i>Fixed-effects</i>							
Provider x Subject	Yes	Yes	Yes	Yes		Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Provider x Subject (CAH3)					Yes		
<i>Fit statistics</i>							
Observations	6,540	6,540	6,540	6,540	19,422	6,540	6,540
R ²	0.979	0.979	0.979	0.979	0.951	0.969	0.974
Within Adjusted R ²	0.023	0.016	0.019	0.024	0.023	0.057	0.090

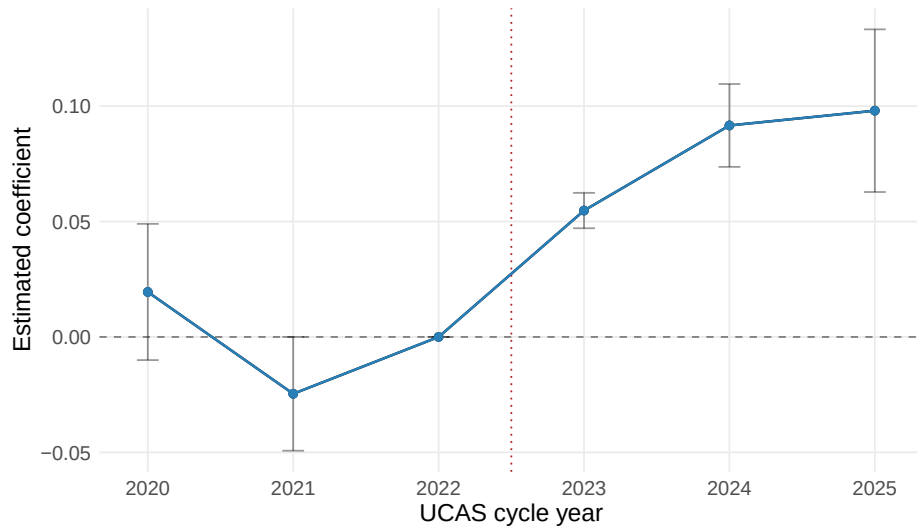
Notes: This table reports robustness checks on the main difference-in-differences estimate. Each column re-estimates the specification from Table 1 under one change. Column 1 repeats the headline model. Columns 2 and 3 replace the treatment with the α index of Eloundou et al. (2024) and the ISCO-2025 index of Gmyrek et al. (2025). Column 4 uses a first-job variant of the exposure index, built from each graduate's first job after university. Column 5 re-estimates the model at the more detailed CAH3 subject level, with standard errors clustered by provider and Revelio field. Column 6 uses the application count in levels as the outcome. Column 7 weights each observation by the degree's application count in the 2022 cycle. All other definitions are as in Table 1. Asterisks denote significance at the 0.01 (***), 0.05 (**), and 0.1 (*) level.

Figure A4: Event study, applications in levels



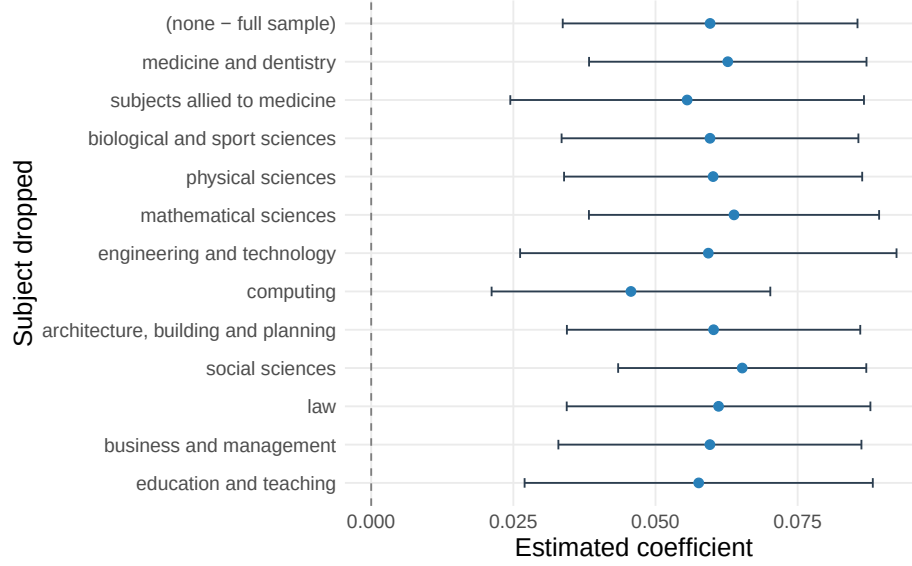
Notes: This figure plots event study estimates from Equation (2), using the number of applications in levels as the outcome. All other definitions are as in Figure 2.

Figure A5: Event study, weighted by 2022 applications



Notes: This figure plots event study estimates from Equation (2), estimated by weighted least squares with each university-subject pair weighted by its application count in the 2022 cycle. All other definitions are as in Figure 2.

Figure A6: Leave-one-subject-out estimates of the main effect



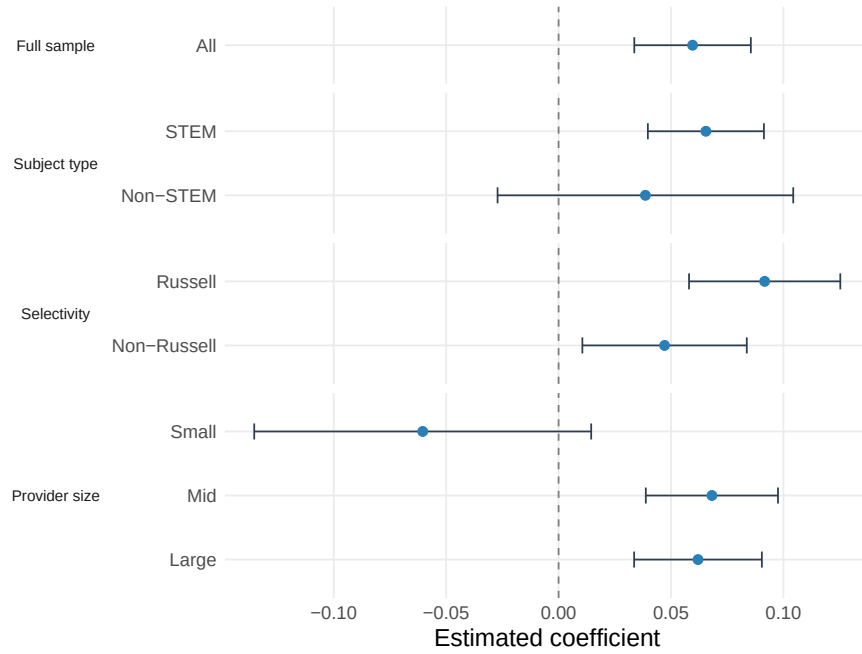
Notes: This figure plots the difference-in-differences estimate from Equation (1), re-estimated dropping one subject group at a time. The top row reports the full-sample estimate. Standard errors are two-way clustered by university and subject. Error bars denote 95 percent confidence intervals.

Table A5: Robustness: university-by-time fixed effects

Dependent variables: Model:	Postings (log)		Junior share ≤ 2	
	(1)	(2)	(3)	(4)
AI exposure x post	-0.0078 (0.0049)	-0.0122** (0.0050)	-0.0025*** (0.0002)	-0.0023*** (0.0001)
<i>Fixed-effects</i>				
University x Subject	Yes	Yes	Yes	Yes
Month	Yes		Yes	
University x Month		Yes		Yes

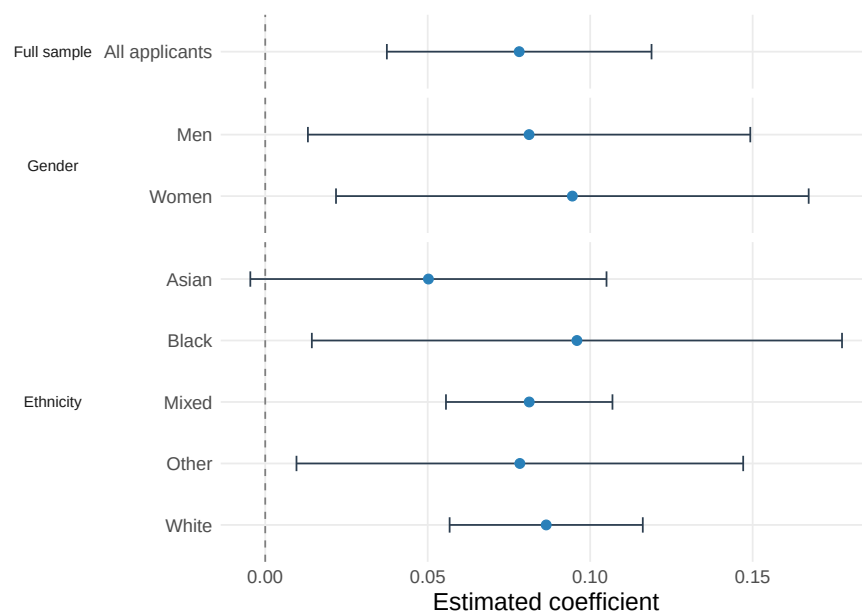
Notes: This table reports difference-in-differences estimates of the effect of AI exposure on log job postings and on the junior share of employment facing each degree's graduates. The junior share is the fraction of positions in the two lowest Revelio seniority levels. The long difference absorbs the university-by-subject and month fixed effects. Columns 2 and 4 add university-by-month fixed effects, which subsume the month effects. Standard errors, in parentheses, are shift-share standard errors that treat occupations as the shocks (Adão et al., 2019). Asterisks denote significance at the 0.01 (***), 0.05 (**), and 0.1 (*) level.

Figure A7: Heterogeneity by subject and university



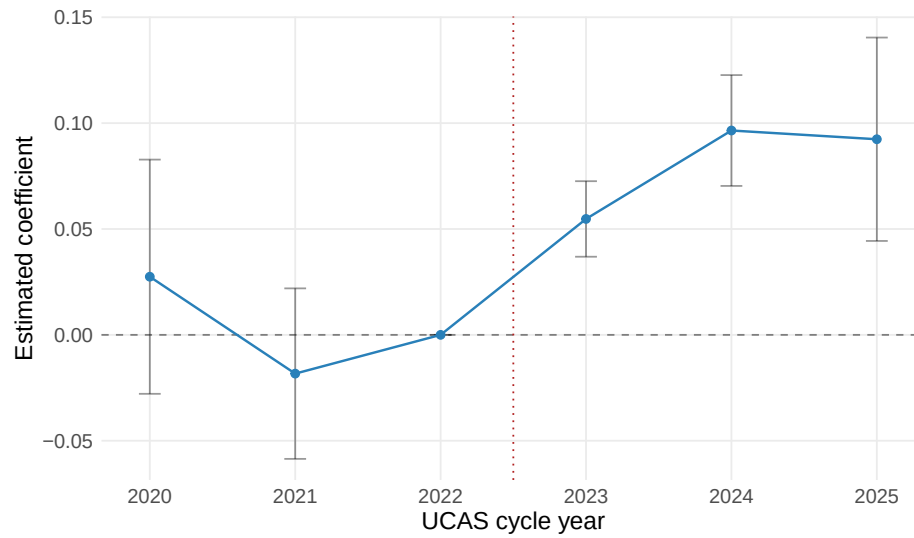
Notes: This figure plots difference-in-differences estimates of the effect of AI exposure on log applications, estimated separately by subject type, university prestige, and university size. Each estimate comes from Equation (1), re-estimated on the indicated subsample, with the full-sample estimate shown as reference. All regressions include university-by-subject and year fixed effects. Standard errors are two-way clustered by university and subject. Error bars denote 95 percent confidence intervals.

Figure A8: Heterogeneity by demographics



Notes: This figure plots difference-in-differences estimates of the effect of AI exposure on log applications, estimated separately by gender and by ethnic group in the public UCAS end-of-cycle data. The treatment is the subject-level exposure index interacted with a post-ChatGPT indicator. All regressions include subject and year fixed effects. Standard errors are clustered by subject. Gender estimates cover UK-domiciled applicants. The ethnicity source does not report domicile, so those estimates cover all applicants. Error bars denote 95 percent confidence intervals.

Figure A9: Event study, public UCAS data



Notes: This figure plots event study estimates of the effect of AI exposure on log national applications by subject, estimated on the public UCAS end-of-cycle data. The treatment is the subject-level exposure index. The regression includes subject and year fixed effects, and standard errors are clustered by subject. All other definitions are as in Figure 2.